

Employee Salary Prediction Using Machine Learning

1. Introduction

The Employee Salary Prediction System is a Machine Learning project designed to estimate employee salaries based on various professional and demographic attributes. The system helps organizations and analysts understand salary trends and make data-driven decisions.

2. Objectives

- Predict employee salary using Machine Learning.
- Perform data cleaning and preprocessing.
- Conduct Exploratory Data Analysis (EDA).
- Compare multiple regression algorithms.
- Deploy the best-performing model using Streamlit.

3. Dataset Description

Dataset Name: employee_salary_dataset_records.csv

Features:

- Age
- Education
- Experience
- Job Role
- Working Hours
- Salary

Total Records: 200

4. Data Preprocessing

The following preprocessing techniques were applied:

- Missing value checking
- Duplicate removal
- Label Encoding
- Feature transformation
- Train-Test Split

5. Exploratory Data Analysis

Performed visualizations:

- Salary Distribution
- Experience vs Salary
- Age vs Salary
- Education Impact Analysis
- Job Role Analysis
- Correlation Heatmap

6. Machine Learning Models

The following models were trained:

Linear Regression

Used as the baseline regression model.

Decision Tree Regressor

Used to capture nonlinear relationships.

Random Forest Regressor

Provided the highest prediction accuracy.

7. Model Evaluation

Evaluation Metrics:

- MAE
- MSE
- RMSE
- R^2 Score

Random Forest Regressor achieved the best overall performance and was selected as the final model.

8. Model Deployment

The trained model was saved as:

salary_model.pkl

A Streamlit web application was developed to allow real-time salary prediction.

9. Additional Enhancements

The following enhancements were implemented beyond project requirements:

- Salary Category Classification
- Career Recommendations
- Download Prediction Report
- Employee Summary Dashboard
- Improved User Interface

10. Conclusion

The Employee Salary Prediction System successfully predicts employee salaries using Machine Learning techniques. The project demonstrates the complete Machine Learning lifecycle including data preprocessing, model training, evaluation, deployment, and user interaction through a Streamlit application.

Technologies Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-Learn
- Joblib
- Streamlit

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